

The Generated Dataset Plottings and Generation Settings

1. Configuration Used

- Channel Model: **3GPP TR 38.901** system-level channel
 - Sionna's **Umi/Uma/Rma**
- Single-link
 - 1 base station antenna
 - 1 user terminal antenna
 - Single polarization
 - Vertical polarization
 - Downlink direction
- Carrier frequency of **3.5 GHz**
- Path loss and shadow fading are **disabled**.
- Time domain waveform **TimeChannel** is used
- Each vector has:
 - **1024 time** samples
 - Bandwidth 1 MHz
 - Maximum delay spread 3 microseconds
- Modulations:
 - BPSK (Custom constellation points)
 - QPSK (Custom constellation points)
 - 16-QAM, 64-QAM, 256-QAM, 1024-QAM (Sionna built-in native QAM Constellations)
- SNR / noise
 - The SNR is generated using Eb/N0 in dB swept from -10 to +20 dB steps of 2 dB (16 points)
 - Converted to noise power through `ebnodb2no()` function of Sionna
 - **No FEC** is used.
- Topology
 - A new random single sector topology of base station + 1 user terminal placement is generated for every batch call

2. Output File Structure

Each three file, namely: Umi.h5, Uma.h5, Rma.h5, contains:

- Data: complex64 array, shape (98304, 1024) = the noisy received I/Q symbol vectors
- Mods: float32 array, shape (98304, 6) = modulation label, order [BPSK, QPSK, 16QAM, 64QAM, 256QAM, 1024QAM]
- SNRs: float32 array, shape (98304,) = the Eb/N0 value (dB) used to generate that row

$98304 = 6 \text{ modulations} * 16 \text{ SNR levels} * 1024 \text{ vectors}$.

Each file is approximately 712-713 MB.

3. Verification Plottings

For each file and SNR level in step of 10 are plotted:

3.1. Rma.h5 plottings

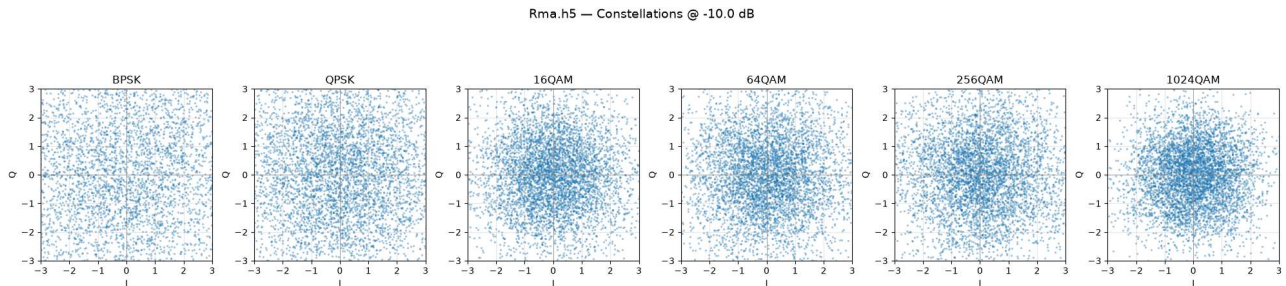


Image 1. Rma.h5 at -10 dB

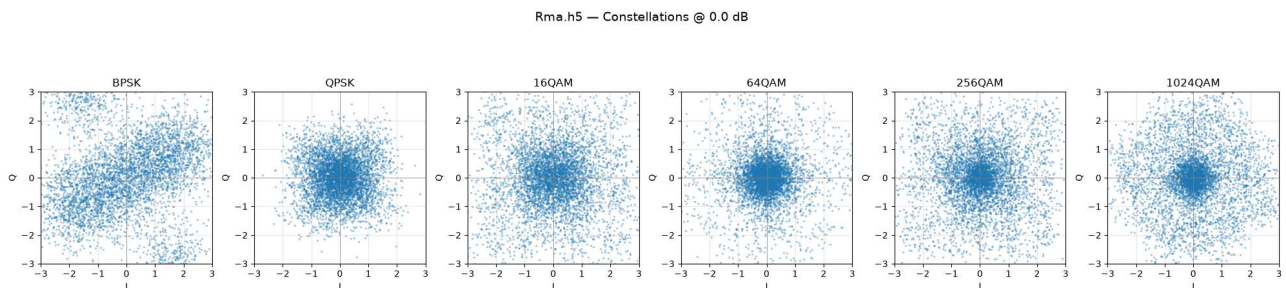


Image 2. Rma.h5 at 0 dB

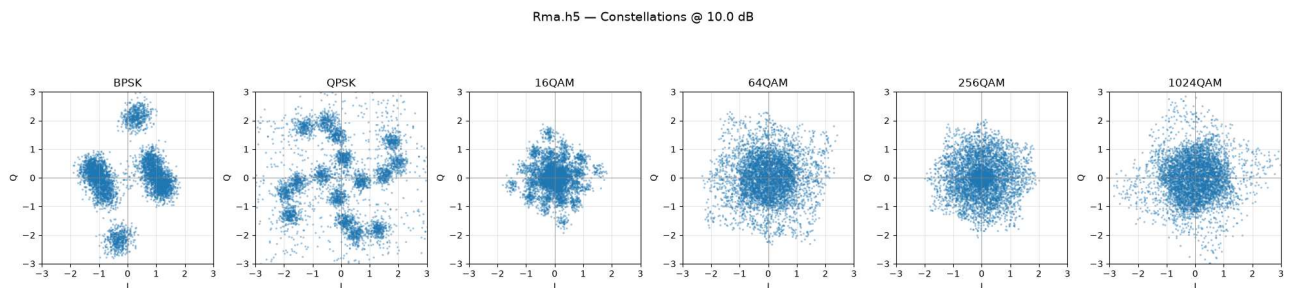


Image 3. Rma.h5 at +10 dB

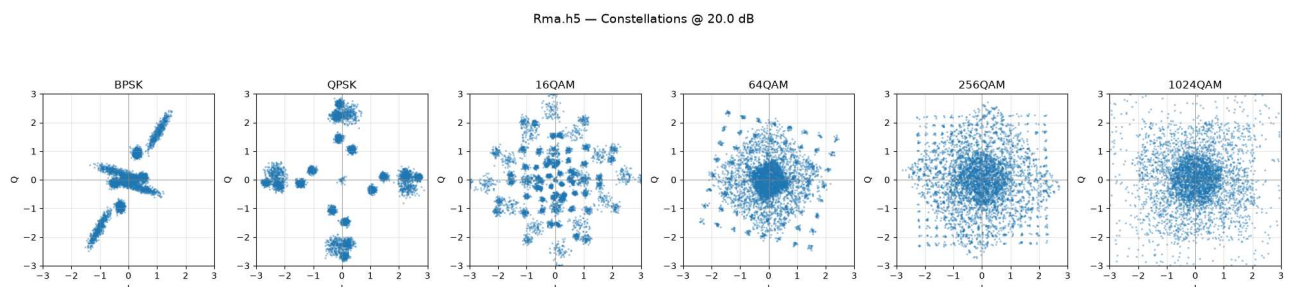


Image 4. Rma.h5 at +20 dB

3.2. Uma.h5 plottings

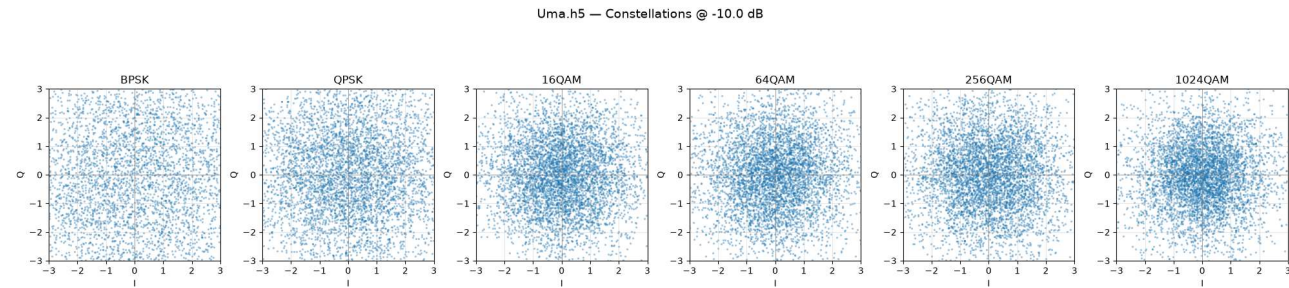


Image 5. Uma.h5 at -10 dB

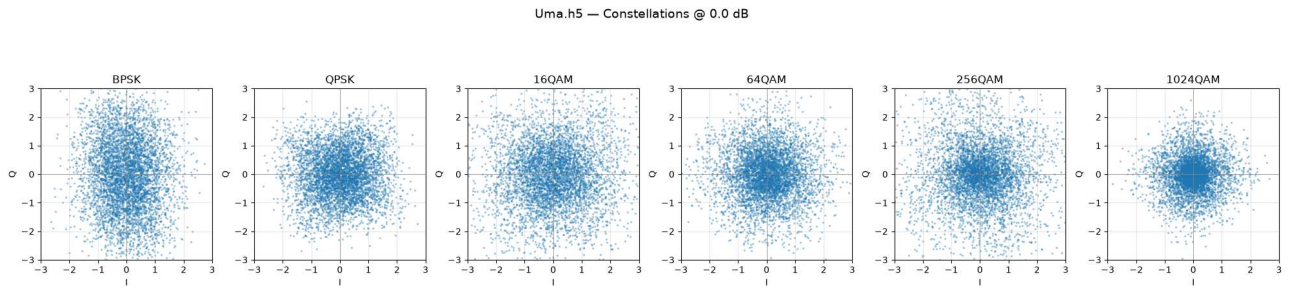


Image 6. Uma.h5 at 0 dB

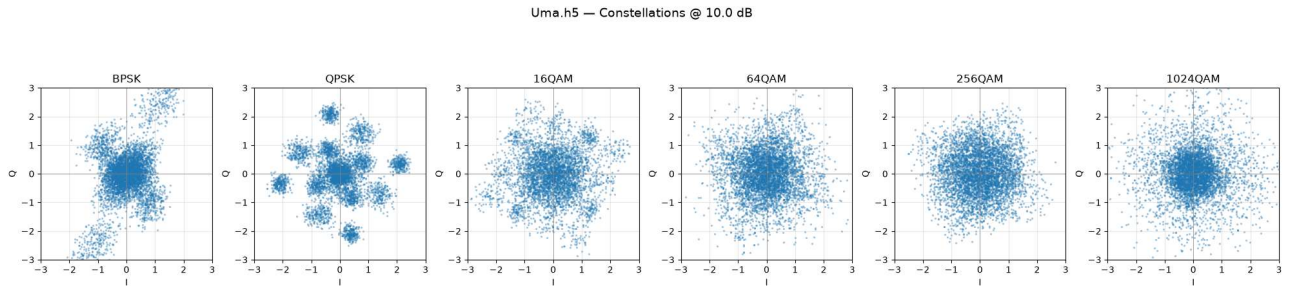


Image 7. Uma.h5 at +10 dB

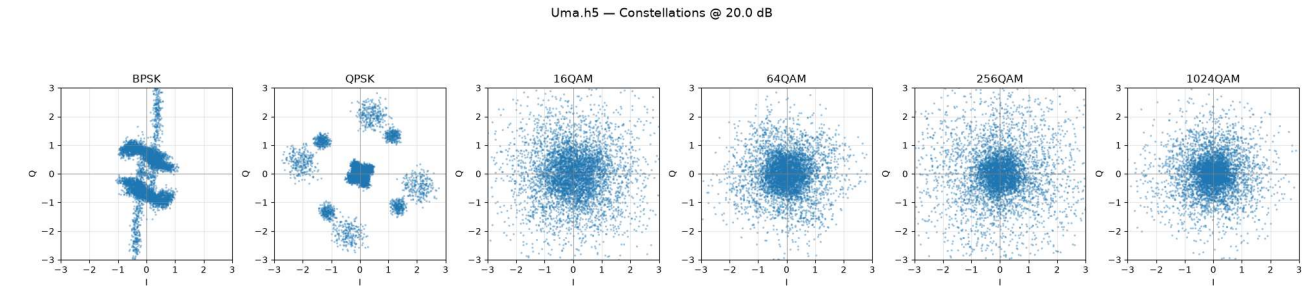


Image 8. Uma.h5 at +20 dB

3.3. Umi.h5 plottings

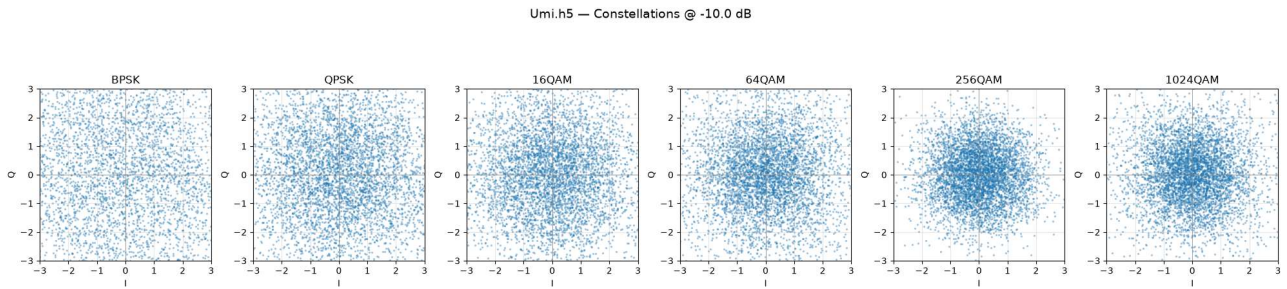


Image 9. Umi.h5 at -10 dB

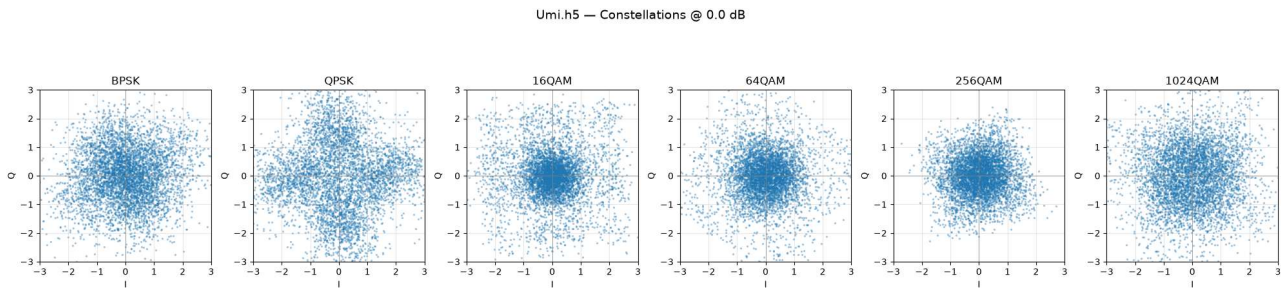


Image 10. Umi.h5 at 0 dB

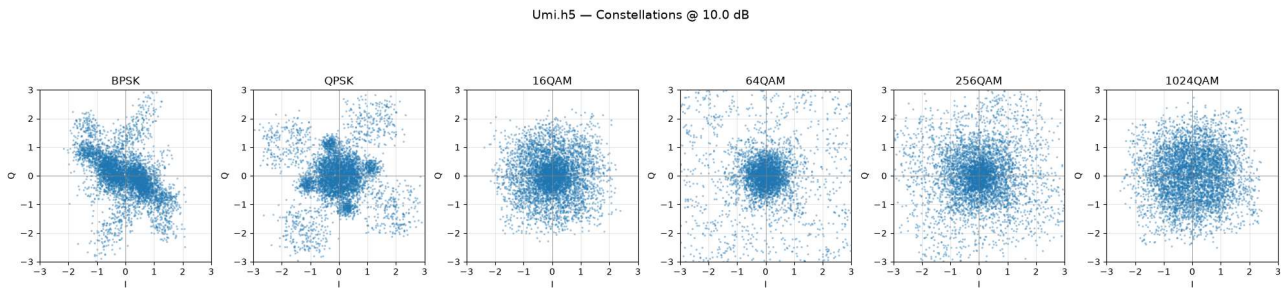


Image 11. Umi.h5 at +10 dB

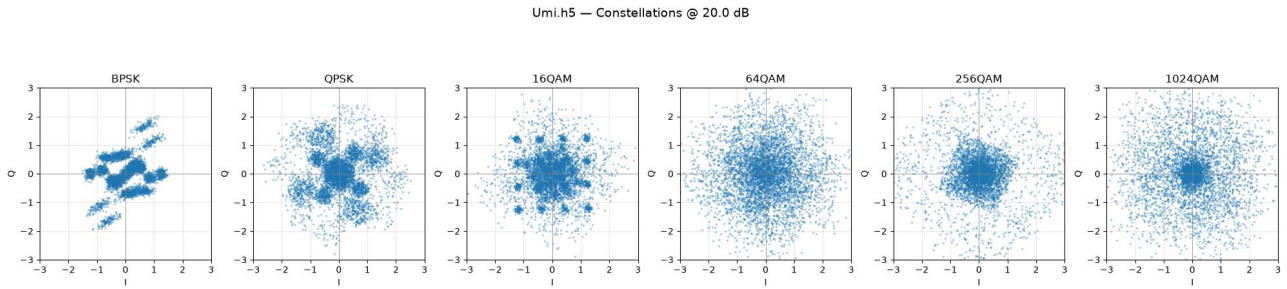


Image 12. Umi.h5 at +20 dB

3.4. Fading behaviour plottings

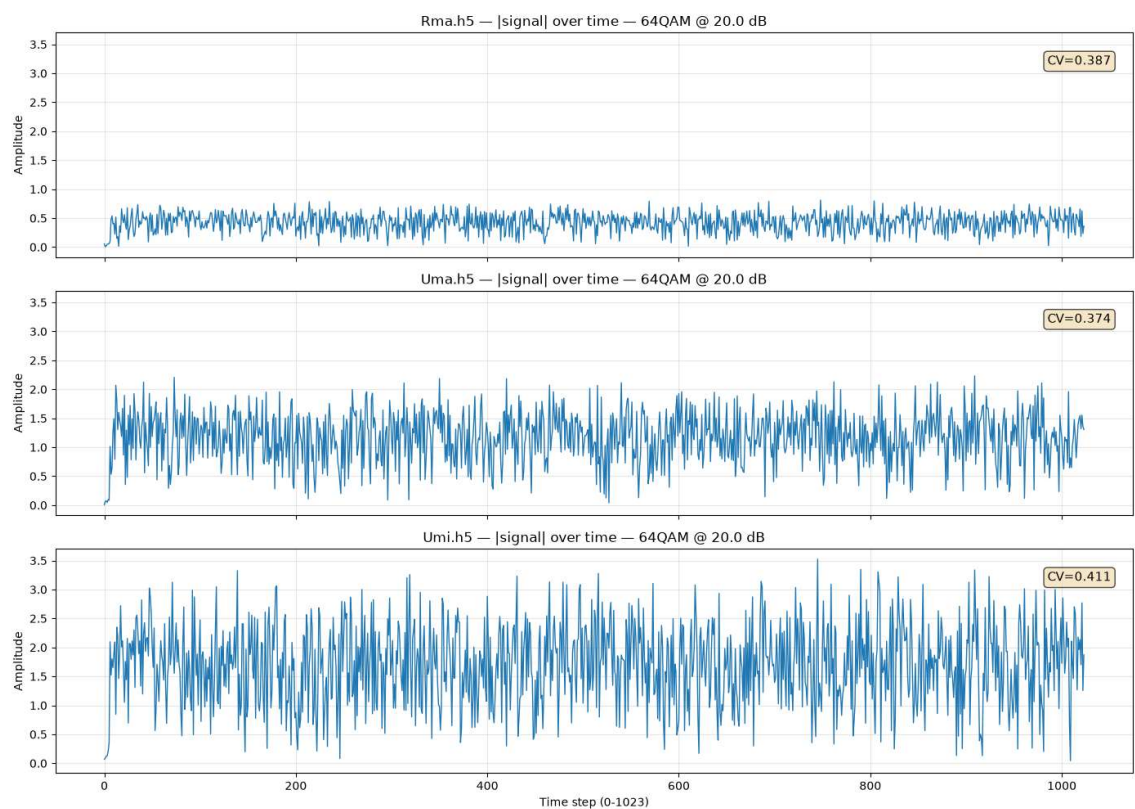


Image 13. Fading Behaviour Comparison

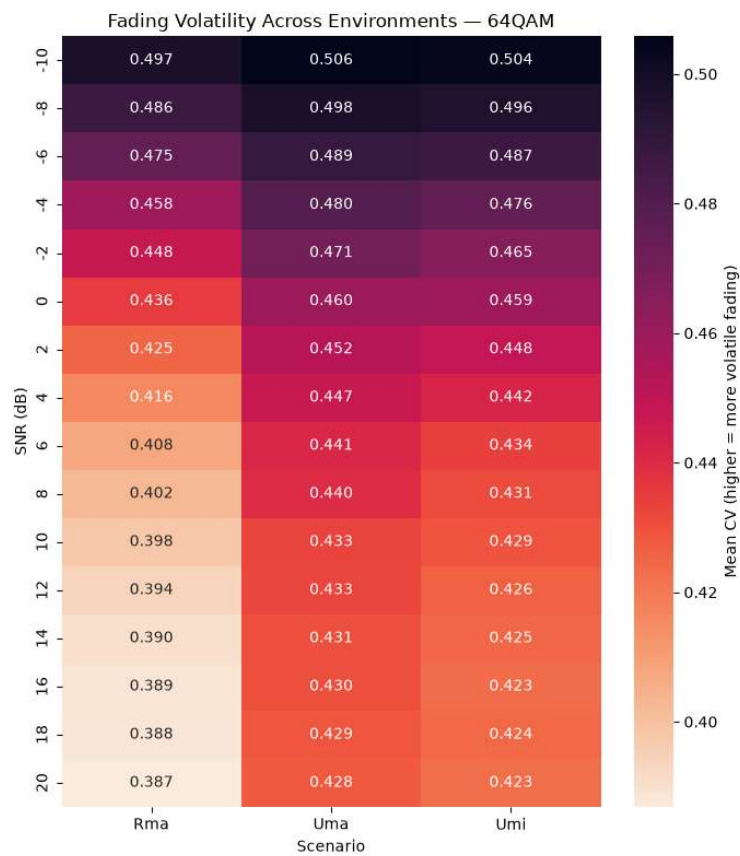


Image 14. Fading Volatility Across Environments for 64-QAM